Notes on Computational Fluid Dynamics: General Principles

2.23 Summary of equations

- **Force** $d\mathbf{f}$ at a surface:

$$\begin{aligned} d\mathbf{f} &= d\mathbf{S}\, p && \text{pressure, Sec. 2.1} \\ d\mathbf{f} &= \mathbf{t}\, dS && \text{traction, Sec. 2.6} \\ d\mathbf{f} &= d\mathbf{S} \bullet \boldsymbol{\sigma} && \text{stress, Sec. 2.6} \end{aligned}$$

- **Rate of deformation**

$$\mathbf{D} = \text{sym}(\nabla \mathbf{u}) \quad \text{Sec. 2.10}$$

Conservation laws

- **Mass**, Sec. 2.4

$$\frac{\partial \rho}{\partial t} + \nabla \bullet (\rho \mathbf{u}) = 0$$

- **Momentum**, Sec. 2.7

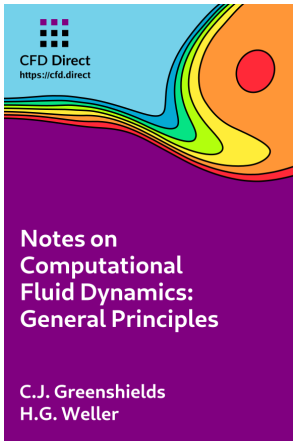
$$\rho \frac{D\mathbf{u}}{Dt} = \nabla \bullet \boldsymbol{\sigma} + \rho \mathbf{b}$$

- **Energy**, Sec. 2.15

$$\rho \frac{De}{Dt} + \rho \frac{DK}{Dt} = -\nabla \bullet \mathbf{q} + \rho r + \nabla \bullet (\boldsymbol{\sigma} \bullet \mathbf{u}) + \rho \mathbf{b} \bullet \mathbf{u}$$

Constitutive models

- **Newtonian fluid**, Sec. 2.12



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$$\boldsymbol{\tau} = 2\mu \operatorname{dev} \mathbf{D} \quad \text{and} \quad \boldsymbol{\sigma} = \boldsymbol{\tau} - p\mathbf{I}$$

- **Fourier's law**, Sec. 2.16

$$\mathbf{q} = -\kappa \nabla T$$

- **Ideal gas**, Sec. 2.16

$$p = \rho R T$$

- **Specific heat capacity**, Sec. 2.18

$$c_V = \left(\frac{\partial e}{\partial T}\right)_V$$

- **Energy-temperature relation**, Sec. 2.19

$$e = \int_{T_{\text{ref}}}^{T} c_V dT + e_{\text{ref}}.$$

Derivatives

- **Material time derivative**, Sec. 2.5

$$\frac{D\boldsymbol{\Psi}}{Dt} = \frac{\partial \boldsymbol{\Psi}}{\partial t} + \mathbf{u} \bullet \nabla \boldsymbol{\Psi}$$

- **Divergence**, Sec. 2.4, $\boldsymbol{\Psi}$ is vector, tensor

$$\nabla \bullet \boldsymbol{\Psi} = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \int_S (d\mathbf{S} \bullet \boldsymbol{\Psi})$$

- **Gradient**, Sec. 2.12

$$\nabla \boldsymbol{\Psi} = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \int_S (d\mathbf{S}\, \boldsymbol{\Psi})$$

- **Laplacian**, Sec. 2.14, $\nabla_{\mathbf{n}} \equiv \mathbf{n} \bullet \nabla$

$$\nabla \bullet (\Gamma \nabla \boldsymbol{\Psi}) = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \int_S (dS\, \Gamma \nabla_{\mathbf{n}} \boldsymbol{\Psi})$$

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